

HiveForce Labs

THREAT ADVISORY



ATTACK REPORT

HybridPetya: A Bootkit-Enabled Ransomware Threat

Date of Publication

September 16, 2025

Admiralty Code

A1

TA Number

TA2025282

Summary

First Seen: February 2025

Targeted Platforms: Windows

Malware: HybridPetya ransomware

Attack: HybridPetya is a new ransomware strain that blends features of Petya and NotPetya while adding modern capabilities. It targets Windows systems by encrypting the Master File Table (MFT), making files inaccessible, and can even bypass UEFI Secure Boot protections through a vulnerability (CVE-2024-7344). Once active, it installs malicious boot components, fakes a “CHKDSK” process, and displays a ransom note demanding Bitcoin for decryption. Although not yet seen in widespread attacks, its Secure Boot bypass makes it a serious emerging threat, highlighting the importance of patches and reliable backups.

CVE

CVE	NAME	AFFECTED PRODUCT	ZERO-DAY	CISA KEV	PATCH
CVE-2024-7344	Howyar Reloader UEFI Application Untrusted Search Path Vulnerability	Howyar Reloader UEFI Application			

Attack Details

#1

HybridPetya is a newly discovered ransomware strain that combines traits of the infamous Petya and NotPetya families while introducing modern capabilities. Like its predecessors, it focuses on crippling Windows systems by encrypting the Master File Table (MFT) of NTFS partitions, which prevents the operating system from locating files. Unlike NotPetya, which was essentially a wiper disguised as ransomware, HybridPetya appears to genuinely support decryption if victims pay the ransom, making it more similar to the original Petya.

#2

HybridPetya is notable for its ability to target modern systems with UEFI support. It installs itself into the EFI System Partition and, in some cases, exploits a vulnerability (CVE-2024-7344) to bypass Secure Boot — a feature designed to stop unauthorized code from running during startup. By doing this, HybridPetya can execute its bootkit even on systems where Secure Boot is enabled, significantly increasing its potential impact on well-protected devices.

#3

When active, HybridPetya first installs its malicious components, forces a reboot, and then runs before Windows loads. It disguises its encryption process as a fake “CHKDSK” disk check to avoid suspicion. Once encryption is finished, the user is presented with a ransom note at boot, demanding payment in Bitcoin and providing a way to enter a decryption key. If the correct key is provided, the bootkit can restore the original bootloader and decrypt the system, allowing normal use again.

#4

So far, HybridPetya has not been observed in widespread attacks. Instead, it has only been found in samples, likely under development or in limited testing. Still, its ability to bypass Secure Boot and affect modern systems makes it a serious potential threat if weaponized. It is not spreading yet, but applying updates and maintaining reliable backups is the safest way to reduce risk.

Recommendations



Apply Security Updates: Patch Windows systems regularly, especially Microsoft’s Secure Boot revocation updates (January 2025 and later), which block the vulnerable UEFI components (CVE-2024-7344). Keep UEFI firmware up to date on all devices.



Harden Boot Security: Ensure Secure Boot is enabled and properly configured. Monitor the EFI System Partition for unauthorized changes (e.g., suspicious files like config, verify, cloak.dat, or replaced bootloaders).



Conduct Regular Data Backups and Test Restoration: Regularly backup critical data and systems, store them securely offline. Test restoration processes to ensure backup integrity and availability. In case of an HybridPetya ransomware attack, up-to-date backups enable recovery without paying the ransom.



Enhance Detection & Monitoring: Deploy endpoint protection solutions capable of detecting ransomware behaviors and bootloader tampering. Use file integrity monitoring on bootloader directories (\EFI\Microsoft\Boot) to detect changes. Watch for unusual reboot events, fake CHKDSK screens, or ransom notes at startup.



Potential MITRE ATT&CK TTPs

<u>TA0005</u> Defense Evasion	<u>TA0008</u> Lateral Movement	<u>TA0040</u> Impact	<u>TA0002</u> Execution
<u>TA0003</u> Persistence	<u>TA0004</u> Privilege Escalation	<u>T1564</u> Hide Artifacts	<u>T1542.001</u> System Firmware
<u>T1587.001</u> Malware	<u>T1587</u> Develop Capabilities	<u>T1587.004</u> Exploits	<u>T1203</u> Exploitation for Client Execution
<u>T1106</u> Native API	<u>T1542</u> Pre-OS Boot	<u>T1059</u> Command and Scripting Interpreter	<u>T1542.003</u> Bootkit
<u>T1574</u> Hijack Execution Flow	<u>T1068</u> Exploitation for Privilege Escalation	<u>T1211</u> Exploitation for Defense Evasion	<u>T1620</u> Reflective Code Loading
<u>T1036</u> Masquerading	<u>T1486</u> Data Encrypted for Impact	<u>T1529</u> System Shutdown/Reboot	<u>T1588</u> Obtain Capabilities
<u>T1027</u> Obfuscated Files or Information	<u>T1564.004</u> NTFS File Attributes		



Indicators of Compromise (IOCs)

TYPE	VALUE
SHA1	BD35908D5A5E9F7E41A61B7AB598AB9A88DB723D, 9DF922D00171AA3C31B75446D700EE567F8D787B, 9B0EE05FFFDA0B16CF9DAAC587CB92BB06D3981B, CDC8CB3D211589202B49A48618B0D90C4D8F86FD, D31F86BA572904192D7476CA376686E76E103D28, A6EBFA062270A321241439E8DF72664CD54EA1BC, C8E3F1BF0B67C83D2A6D9E594DE8067F0378E6C5, C7C270F9D3AE80EC5E8926A3CD1FB5C9D208F1DC, 3393A8C258239D6802553FD1CCE397E18FA285A1, 98C3E659A903E74D2EE398464D3A5109E92BD9A9, D0BD283133A80B47137562F2AAAB740FA15E6441
File names	bootmgfw.efi, core.dll, f20000.mbam_update.exe, improved_not_petyanew.exe, notpetya_new.exe, notpetyanew.exe, notpetyanew_improved_final.exe, bootmgfw.efi, cloak.dat

Patch Links

<https://uefi.org/revocationlistfile>

<https://msrc.microsoft.com/update-guide/vulnerability/CVE-2024-7344>

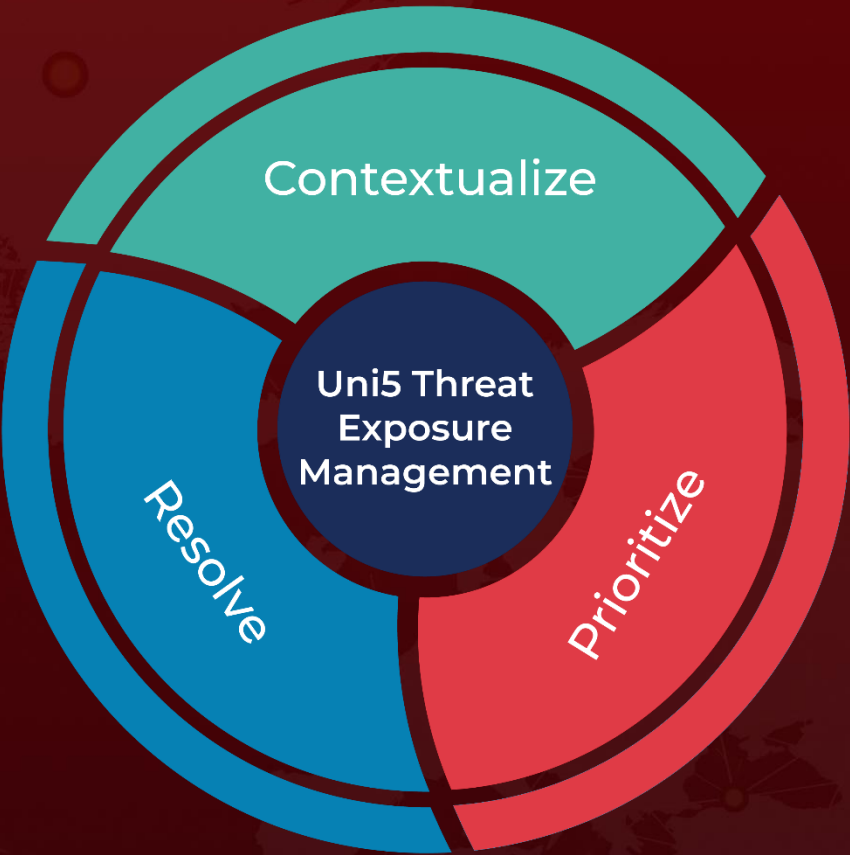
References

<https://www.welivesecurity.com/en/eset-research/introducing-hybridpetya-petya-notpetya-copycat-uefi-secure-boot-bypass/>

What Next?

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